Revisiting Geopolitical Alignment: Political and Economic Implications

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PRELIMINARY DRAFT - DO NOT CIRCULATE

Abstract

In a multipolar global order, properly measuring countries' geopolitical alignment is an important research objective. While using UNGA voting patterns as a proxy for alignment is well established in the literature, I posit that bilateral geopolitical distance does not take into account the reciprocal nature of international relations: how country i regards country j need not coincide with how j regards i. I propose a refined, rank-based measurement, the Alignment Percentile, which is comparable across years and is not symmetric within a country pair. The new measure delivers three results. First, it reproduces the diplomatic history of the Cold War and of the USA–China rivalry more faithfully than the distance-based index in current use, which mistakes movements of the hegemons for movements of third countries. Second, it shows that Geoeconomic Vulnerability has two faces: a country's own geopolitical ranking of its trading partners and its partners' ranking of it. Third, geopolitical reciprocity enters the gravity equation with a sign that changes over time: unbalanced pairs traded less in the 1990s and trade more today.

1 Introduction

The United Nations were founded in the aftermath of World War II, its main goal being the development of a new international order based on collegiality and representation of all nations.[1] At the core of the UN lies its General Assembly, where votes on non-binding resolutions allow countries to signal their position on international issues such as conflicts or humanitarian crises. Researchers in International Relations have worked out a number of statistical procedures to analyse patterns of UNGA voting.[2] The most successful development in this

[1] Not all countries are admitted in the UN and, especially at its 1945 inception, it was a fairly exclusive organisation. See e.g. Hanhimäki (2015) and Luard (1982).

[2] See the discussion of the UN voting data literature in Bailey, Strezhnev, and Voeten (2017), and Kim and Russett (1996) and Voeten (2000) for the voting blocs that emerged after the Cold War.

field has been Bailey, Strezhnev, and Voeten (2017), wherein the authors employ a spatial ordinal model and compute a distribution of countries' ideal points by year. By taking the Ideal Point Distance (henceforth IPD) between any pair of countries, researchers can use these ideal point estimates to proxy for the geopolitical distance between countries. In the nascent field of Geoeconomics, IPD has been used in relation to the impact it might have on economic relations between countries, such as trade flows and FDI.[3] Another relevant application has consisted in estimating countries' relative geoeconomic positioning. In this particular framework, the discussion has focused on the emergence of connector countries.[4]

This paper argues that the current use of IPD is conceptually flawed and proposes an alternative methodology that provides new perspectives on Geoeconomics. The argument rests on two observations. The first is that a distance is a cardinal object measured on a scale that has no natural unit: the same IPD can describe a close ally in a year in which the Assembly is polarised and a lukewarm partner in a year in which it is not. The second is that a distance is symmetric by construction, whereas international relations are not. The asymmetry of political and economic ties has been a central theme of International Relations at least since Hirschman (1945), who showed how unequal dependence turns trade into an instrument of national power, and it underlies both the study of reciprocity as a foundation of cooperation (Keohane 1986) and the recent work on weaponised interdependence and economic coercion (Farrell and Newman 2019; Clayton, Maggiori, and Schreger 2024). A measure of alignment that returns the same number for the way Washington regards Ankara and for the way Ankara regards Washington cannot speak to any of these questions.

I therefore replace distances with ranks. For every reference country and year I order all other UN members by their IPD from the reference country and express each country's position as a percentile, the Alignment Percentile (AP). The percentile states where a country stands in the reference country's ordering of the world. It is comparable across years, it is unaffected by rescaling of the ideal-point dimension, and it is not symmetric: the position of i in the ordering of j generally differs from the position of j in the ordering of i.

The contribution of this paper is threefold.

First, I propose a quantitative assessment of countries' geopolitical alignment during the Cold War and the current USA–China hegemonic rivalry (Section 4.1). For five countries with very different strategic trajectories, I compare an index built on AP with the distance-based index of Airaudo et al. (2025) and confront both with the diplomatic record. The percentile-based index tracks documented turning points, such as the Turkish–American crisis over Cyprus, the Gaullist decade, and India's oscillations between Moscow and Washington, and it does not produce the implausible readings of the distance-based index, according to which the United Kingdom and France drifted away from the United States under Thatcher and Mitterrand, Türkiye has stood closer to China than

[3] See Gopinath et al. (2025).

[4] See Aiyar and Ohnsorge (2024), Airaudo et al. (2025), Baur et al. (2025).

to the United States for a quarter of a century, and India has been almost perfectly aligned with China.

Second, I show that current measures of Geoeconomic Vulnerability (GeoV) are not consistent across time and lead to erroneous interpretations that can be remedied by integrating my proposed measure (Section 4.2). Because AP is not symmetric, vulnerability splits into two components: the distance at which a country places its trading partners, and the distance at which its trading partners place it. The first rose for four decades and collapsed after 1990; the second barely moved. The aggregate hides a persistent hierarchy, in which the Western bloc is the least exposed on every measure, and one important exception, the United States, whose partners have ranked it as distant since the 1990s.

Third, a new empirical puzzle on the relationship between geopolitical reciprocity and trade is introduced (Section 4.3). In a structural gravity model, unbalanced relationships traded significantly less than balanced ones in the 1990s and early 2000s; the penalty has since disappeared and, after 2022, it has turned into a sizeable premium.

These results matter beyond measurement. If the index used to classify countries as aligned, non-aligned or connectors confounds the movement of a hegemon with the movement of everybody else, then the fragmentation scenarios, the friend-shoring strategies and the vulnerability rankings built on that classification inherit the error. The evidence presented here suggests that several countries currently described as drifting between blocs have not moved at all, that exposure to geopolitically distant partners is an old and declining feature of world trade and not a new one, and that the mutual or one-sided nature of a political relationship carries information for trade that a symmetric distance cannot convey.

The remainder of the paper is organised as follows. Section 2 describes the data, Section 3 reviews the literature, Section 4 presents the measures and the empirical results, and Section 5 concludes.

2 Data

To measure geopolitical distance, I use data from Bailey, Strezhnev, and Voeten (2017). My attention focuses on the following countries: the United Kingdom, Türkiye, France, India and Yugoslavia. The choice of this sample is justified by the fact that these countries have played heterogeneous roles in their strategic choices vis-à-vis the USA and USSR: two Western permanent members of the Security Council with a colonial legacy and, in the French case, an explicit claim to strategic autonomy; a NATO member on the Soviet border whose loyalty to the alliance was repeatedly tested; and the two leading founders of the Non-Aligned Movement, one of them socialist. Such differences, which will be discussed in depth in the results, are instrumental in understanding the nuances captured by the new measure here proposed.

Two periods are considered. The USA–USSR rivalry is studied from the first

sessions of the General Assembly to 1990. The USA–China rivalry is studied from 1991 to 2024; an earlier start would be of limited meaning, since the seat of China was held by the government in Taipei until Resolution 2758 of October 1971. Yugoslavia leaves the sample at its dissolution in 1991–92 and is therefore analysed only for the Cold War.

For data on International Trade, I integrate the historical dataset from Fouquin and Hugot (2016) with USSR and Yugoslavia data from Campos, Heid, and Timini (2024) and with Gaulier and Zignago (2010). The resulting panel of bilateral flows covers the post-war period up to 2024.

3 Literature Review

A first strand of literature examines the measurement of geopolitical preferences and international alignment through voting behaviour in the UNGA. Among the most influential contributions, Bailey, Strezhnev, and Voeten (2017) develop a dynamic spatial ordinal model to estimate countries' ideal points over time, providing a systematic framework for measuring political preferences and their evolution over time. The resulting ideal points have subsequently been used to construct bilateral measures of geopolitical distance, generally interpreted as proxies for the degree of political affinity between countries. More broadly, the literature has employed treaty commitments and international agreements as alternative indicators of international alignment (Simmons and Hopkins 2005; Lupu 2013; Broner, Martin, Meyer, Trebesch, and Zhou Wu 2025). Recent contributions have also explored the relationship between geopolitical alignment and the structure of the international economic order, emphasizing how shifts in global hegemony and political preferences shape international integration (Broner, Martin, Meyer, and Trebesch 2025; Broner, Martin, Meyer, Trebesch, and Zhou Wu 2025). Despite these advances, existing approaches largely conceptualize geopolitical alignment through the distance between countries' political preferences, leaving open the question of whether such measures adequately capture the reciprocal nature of bilateral geopolitical relationships.

A second strand of literature investigates the economic consequences of geopolitical alignment and the emergence of geoeconomic fragmentation. That trade follows political alignment is an old finding of International Relations: Pollins (1989) shows that bilateral flows respond to the state of diplomatic relations, and Gowa and Mansfield (1993) that trade is larger within alliances than across them. Recent research documents how geopolitical tensions increasingly influence international trade patterns, investment decisions, and the organization of global value chains. Gopinath et al. (2025) examine changes in global economic linkages and the extent to which trade relationships are becoming increasingly structured along geopolitical lines. Similarly, Bonadio et al. (2024) quantify the potential economic consequences of decoupling between geopolitical blocs, while Campos, Heid, and Timini (2024) provide historical evidence on the trade consequences of geopolitical fragmentation during the Cold War and Gokmen (2017) shows that Cold War blocs dominated other determinants

of trade until 1990. From a broader theoretical perspective, Clayton, Maggiori, and Schreger (2024) emphasize how economic interdependence can become a source of geopolitical leverage, as dominant countries may exploit asymmetric trade and financial relationships to pursue strategic objectives, an argument that formalises the insight of Hirschman (1945). This mechanism is closely related to the concept of geoeconomic vulnerability, whereby countries' dependence on specific economic partners may expose them to geopolitical pressure. In this context, Aiyar and Ohnsorge (2024) investigate the emergence of connector countries, while Baur et al. (2025) analyse the role of non-aligned economies in mitigating the welfare costs of fragmentation. These contributions highlight the importance of understanding how geopolitical positioning translates into economic exposure, although the measurement of such exposure remains closely connected to the underlying definition of geopolitical alignment.

Finally, an important methodological issue concerns how geopolitical distance is constructed and interpreted in empirical applications. The use of ideal point distances implicitly assumes that the relative position of two countries in a common ideological space provides a sufficient representation of their bilateral geopolitical relationship. However, distance is inherently symmetric: the ideological distance between country i and country j is identical to that between j and i. This property may obscure relevant asymmetries in international relations, particularly when countries occupy different positions in the global distribution of political power. The literature on geoeconomic fragmentation increasingly emphasizes the importance of these asymmetric relationships, particularly in the context of hegemonic competition and economic coercion (Clayton, Maggiori, and Schreger 2024; Broner, Martin, Meyer, and Trebesch 2025), and the International Relations literature has long treated reciprocity, understood as the exchange of roughly equivalent treatment, as a condition for stable cooperation (Keohane 1986). Moreover, alternative approaches to measuring geopolitical distance and alignment continue to be debated, as illustrated by Airaudo et al. (2025). This paper contributes to this debate by questioning whether conventional ideal point distances adequately capture bilateral geopolitical alignment and by introducing an explicit measure of reciprocity. It then investigates the implications of this distinction for the measurement of geoeconomic vulnerability and for the empirical relationship between geopolitical alignment and bilateral trade.

4 Empirics

4.1 Alignment Measurements

Assuming competition between two hegemons, a natural question is how equidistant other countries are from these two powers. Airaudo et al. (2025) propose the following measure: given a country i and two poles j and k in year t,

$$seg_{ijkt} = \frac{IPD_{ikt} - IPD_{ijt}}{IPD_{ijt} + IPD_{ikt}}. \tag{1}$$

Here, IPD_{ijt} denotes the Ideal Point Distance between countries i and j, estimated by Bailey, Strezhnev, and Voeten (2017) using UNGA voting data. In what follows I refer to (1) as the Alignment Index; it is labelled SEG in the figures.

By construction, seg_{ijkt} is bounded between -1 and 1. A value of 1 indicates that country i is perfectly aligned with country j, whereas a value of -1 indicates perfect alignment with country k. A value of zero implies that i is equidistant from j and k.

Despite its conceptual appeal and straightforward interpretation, seg_{ijkt} has several shortcomings as an indicator:

- It only indicates whether country i is equally distant from hegemons j and k, but does not establish whether i is actually close to either of them.
- Changes in the coefficient may reflect changes in the absolute distance from one pole, even if the distance from the other remains approximately constant. Such changes may therefore reflect movements in the ideal points of the hegemons themselves rather than a genuine shift in the relative geopolitical positioning of country i.
- It does not account for the relative position of other countries with respect to j and k. In other words, it is insensitive to the distribution of ideal points in a given year.
- In a geopolitical context involving more than two hegemons, it is not possible to compute a meaningful measure of equidistance using one-dimensional data such as ideal points.

The second and third shortcomings are not hypothetical. The two superpowers of the Cold War changed their own position in the Assembly considerably. In the 1970s and 1980s the United States found itself increasingly isolated from a majority enlarged by decolonisation, on questions ranging from the Middle East and southern Africa to the New International Economic Order (Moynihan and Weaver 1978; Westad 2005); after 1985, Soviet diplomacy under Gorbachev moved towards that majority (Gaddis 2005). Both movements mechanically lengthen every country's distance from Washington and shorten every country's distance from Moscow, and (1) records them as if all third countries had moved towards the USSR. Likewise, a pole that sits in the crowded centre of the distribution, as China does today among the developing countries, is close in absolute terms to a very large number of states, most of which will appear aligned with it under (1) whatever their actual foreign policy.

I therefore propose an alternative measure, constructed as follows:

- Given a reference country j (for instance a hegemon) and a country i in year t, compute IPD_{ijt} for all countries $i \neq j$ voting in the UNGA in year t.

- Rank countries in descending order of IPD_{ijt}, assigning a rank of 1 to the country that is geopolitically furthest from j, a rank of 2 to the second furthest, and so on.
- Divide each country i's rank by the total number of countries voting in the UNGA in year t.

The resulting measure, which I call the Alignment Percentile (AP_{jit} henceforth), is the percentile occupied by country i in the ordering of the world held by country j in year t: the first subscript denotes the country whose ordering is used, the second the country that is being ranked. A higher (resp. lower) AP_{jit} indicates greater (resp. smaller) relative geopolitical proximity of i to j.

AP_{jit} addresses the shortcomings of seg_{ijkt}. It is an absolute statement about closeness to a given pole, because it locates i among all countries and not only relative to the other pole. It is invariant to any change in the hegemon's own position that leaves the ordering of the other countries unchanged, and to any stretching or compression of the ideal-point scale. It depends on the whole distribution of ideal points in a year, so that being close to a pole means being closer than most other countries are. And since it is defined pole by pole, it can be computed for any number of hegemons.

Two further properties are worth stressing. The first is that AP is not symmetric: in general $AP_{jit} \neq AP_{ijt}$. A country that sits at one end of the ideal-point line, as the United States has done for most of the UN's history, is ranked among the most distant by most other members, while its own ordering of the world spreads the other members over the whole range. A moderate country may therefore be in the closer half of Washington's ordering while Washington is in the most distant decile of its own. This property is exploited in Sections 4.2 and 4.3. The second is a cost: by construction a rank discards cardinal information, so that AP is silent on how much further the country ranked tenth is than the country ranked ninth, and it is sensitive to changes in UN membership, which add or remove entries from every ordering.

I illustrate the properties of this new measure by comparing it with the conventional Alignment Index. I choose two prominent examples of geopolitical rivalry: the USA–USSR competition during the Cold War and the contemporary USA–China rivalry. In particular, I compute

$$HI_{ijkt} = \frac{AP_{jit} - AP_{kit}}{AP_{jit} + AP_{kit}}, \tag{2}$$

which I call the Hegemon Index.

HI_{ijkt} is, by construction, bounded in the interval $[-1; 1]$. A value approaching 1 (resp. -1) implies that i is among the countries closest to j (resp. k) and among the most distant from k (resp. j). In all figures of this section the first pole j is the USA, so that positive values (in red) denote a position closer to the USA and negative values (in blue) a position closer to the USSR or China; solid lines report the Alignment Index and dashed lines the Hegemon Index.

4.1.1 Cold War

In this section, I examine the alignment behaviour of five countries over the course of the Cold War: Türkiye, the United Kingdom, India, France and Yugoslavia. The rationale behind this selection is to compare countries with different geopolitical orientations and identify common patterns and divergences in their voting behaviour. Figures 1–5 report the two indices for each country, and Table 1 at the end of the section summarises how each of them compares with the historical record.

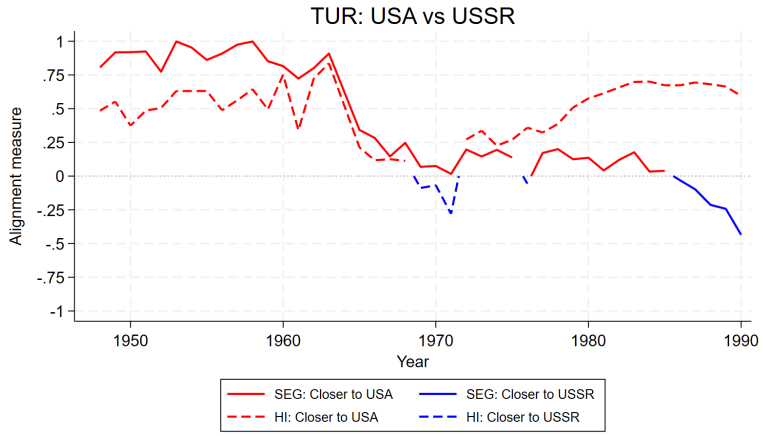


Figure 1: Türkiye: HI vs SEG during the Cold War.

Türkiye is often mentioned as a paradigmatic example of a middle power.[5] Having sent troops to Korea in 1950 and joined NATO in 1952, it enters the sample firmly in the Western camp, and both indices in Figure 1 agree on this until 1963, although the Alignment Index, close to one, overstates a proximity that the HI places mostly between 0.4 and 0.65. According to both series, the beginning of Türkiye's more independent geopolitical positioning can be traced to the first half of the 1960s. The Cuban Missile Crisis, which ended with the removal of the Jupiter missiles from Turkish soil, made the Turkish government particularly concerned about relying unconditionally on the American defence system, as well as about the possibility of being used as a bargaining chip by the USA.[6] The sharp fall of both indices after 1963 coincides more precisely with the Cyprus crisis and with President Johnson's letter of June 1964, which warned Ankara that NATO might not come to its defence if an intervention on the island provoked a Soviet reaction; the following years saw a deliberate normalisation of relations with Moscow, culminating in the Soviet–Turkish economic cooperation agreement of 1967 (Hale 2012). The HI turns negative between 1969 and 1971

[5] See, for instance, Oğuzlu (2023) and Gulmez (2020).
[6] See Hale (2012), page 110.

and touches zero again in 1976, in the middle of the arms embargo imposed by the US Congress after the 1974 Turkish intervention in Cyprus.

The two indices part ways in what follows. The HI shows that Türkiye's diplomatic divergence from the United States, accompanied by a relative rapprochement with the USSR, was rather short-lived and lasted for approximately one decade: once the embargo was lifted in 1978, and even more so after the defence cooperation agreement and the military coup of 1980, the HI climbs back to about 0.7, above the level of the 1950s. The Alignment Index instead stays close to zero for the rest of the Cold War and turns negative after 1985, which would make the Türkiye of Turgut Özal closer to Moscow than to Washington. No account of Turkish foreign policy supports this reading. The decline is the footprint of the movements of the two superpowers described above and not of a movement of Ankara.

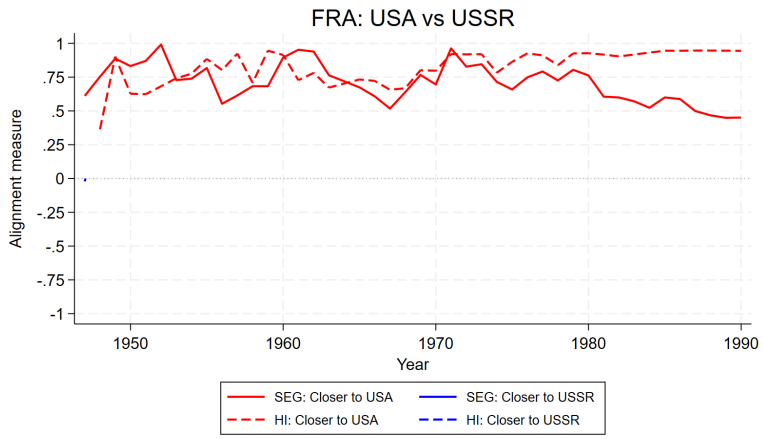


Figure 2: France: HI vs SEG during the Cold War.

France and the USA have been close allies since the American War of Independence. Nevertheless, their diplomatic relations have been characterised by diverging perspectives on several issues, including the role of NATO in Europe, the Vietnam War and the Suez Crisis.[7] Figure 2 registers these fluctuations. The HI starts from low values in 1947–48 and jumps to about 0.9 in 1949, the year of the North Atlantic Treaty. Both indices then record the Gaullist decade: after de Gaulle's return to power in 1958, the recognition of the People's Republic of China in 1964, the withdrawal from NATO's integrated military command in 1966 and the open criticism of the American war in Vietnam, the HI settles between 0.65 and 0.8 and reaches its minimum in 1967–68, while the Alignment Index falls to about 0.5 in 1967 (Bozo 2001). The Alignment Index also displays a marked trough in 1956, the year in which Washington sided against Paris and London over Suez; the HI is less sensitive to that episode, which says that France

[7] See Statler (2016).

remained among the countries closest to the USA even while disagreeing with it.

The subsequent rapprochement is where the HI is more informative. From 1971 onwards the HI stays around or above 0.9, with brief dips in the mid-1970s, and is essentially flat in the 1980s, the years in which Mitterrand supported the deployment of American intermediate-range missiles in Europe. The Alignment Index declines instead from about 0.8 in 1979 to 0.45 in 1990, the lowest value of the whole period, below the level of the Gaullist years. An index according to which France was more distant from the United States in 1990 than in 1967 fails a basic test of historical plausibility.

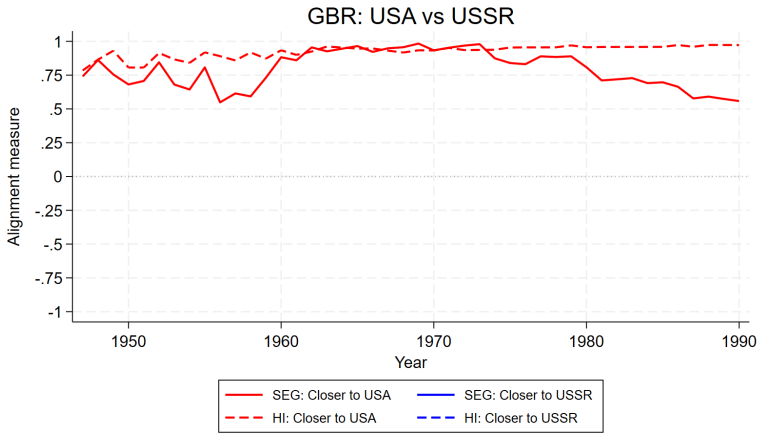


Figure 3: United Kingdom: HI vs SEG during the Cold War.

According to the Alignment Index in Figure 3, the United Kingdom and the USA diverged considerably in their UNGA voting patterns until the end of the 1950s, with a minimum in 1956, the year of Suez. This is the period in which colonial questions dominated the agenda of the Assembly and in which Washington repeatedly distanced itself from the European colonial powers (Louis and Robinson 1994). The HI is much less affected and remains between 0.8 and 0.93: the two countries disagreed, but the United Kingdom never ceased to be one of the members closest to the United States and never moved towards the USSR.

With the acceleration of decolonisation after 1960, the two indices converge at values above 0.9.

While the Alignment Index suggests a gradual shift towards the USSR during the second half of the Cold War, falling from about 0.9 in the late 1970s to 0.55 in 1990, the HI tells a different story. Particularly from 1960 onwards, the United Kingdom consistently ranked among the countries most closely aligned with the USA and most distant from the USSR, and the HI reaches its maximum, about 0.97, in the second half of the 1980s. The latter reading is the one consistent with the history of the Anglo-American relationship, which was at its closest in

the Thatcher–Reagan years (Dumbrell 2006). The United Kingdom and France thus display the same artefact: the Alignment Index falls in the 1980s for both, by a similar amount and for the same reason.

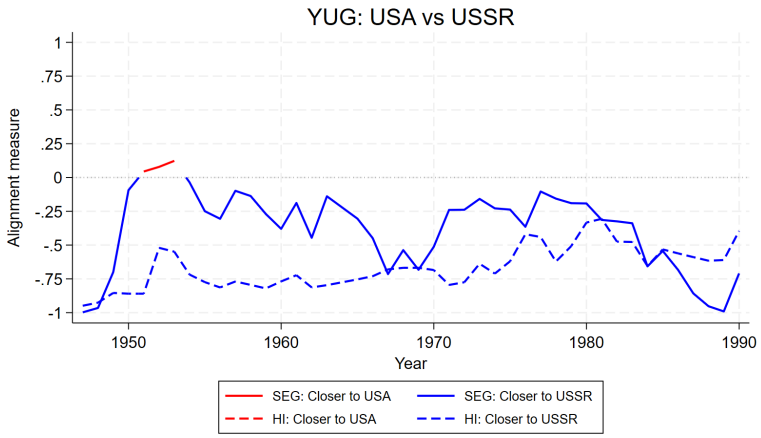


Figure 4: Yugoslavia: HI vs SEG during the Cold War.

Yugoslavia was a founding member of the Non-Aligned Movement. Despite its socialist institutional setting, it sought to detach itself from the direct influence of the USSR after the Tito–Stalin split of 1948. The Alignment Index in Figure 4 suggests a degree of geopolitical proximity to both superpowers: it rises from -1 in 1947 to positive values in 1951–53, the years of American military assistance and of the Balkan Pact with Greece and Türkiye (Lees 1997), and fluctuates between -0.1 and -0.45 for most of the following two decades. However, the HI reveals that Yugoslavia consistently voted relatively close to the USSR and far from the USA throughout the Cold War. The Western tilt of the early 1950s is visible, as the HI rises from -0.85 to about -0.5, and so is its reversal after Khrushchev's visit to Belgrade in 1955; but at no point does Yugoslavia leave the half of the Assembly closer to Moscow. A slow movement towards the centre takes place in the 1970s and peaks around 1980–81, when Belgrade condemned the Soviet intervention in Afghanistan.

This result should be read with care. Being closer to the USSR in the Assembly does not mean being a Soviet satellite. On the questions that filled the agenda, namely decolonisation, apartheid, disarmament and the economic demands of the developing world, the positions of the Non-Aligned Movement and those of the Soviet bloc largely coincided, also because Moscow courted the new majority (Westad 2005; Dinkel 2019). What the HI shows is that non-alignment, as practised in New York, was not equidistance.

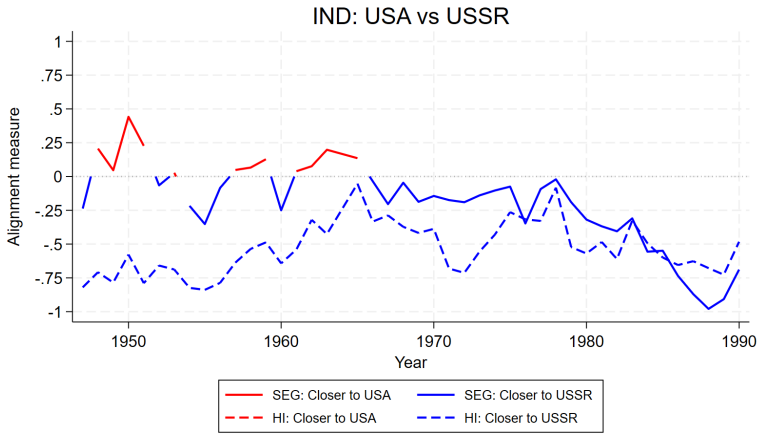


Figure 5: India: HI vs SEG during the Cold War.

India was the largest country among the members of the Non-Aligned Movement. It never formally aligned itself with either the USA or the USSR. According to the Alignment Index in Figure 5, the former British colony remained roughly equidistant from the two superpowers until the mid-1970s, and was even closer to the USA in several years before 1965. However, the HI reveals that India consistently voted closer to the USSR than to the USA. This suggests a less neutral diplomatic position than the one inferred from the Alignment Index and commonly discussed in the literature.[8]

More importantly, the movements of the HI follow the known phases of Indian foreign policy closely. The index is at its most negative, about -0.8, in 1954–55, when the United States concluded its military pact with Pakistan and Khrushchev and Bulganin visited India (McMahon 1994). It then rises steadily and comes close to zero in 1965, after the war with China of 1962 had brought American and British military assistance. It falls again after the Indo-Pakistani war of 1965 and reaches about -0.7 in 1971–72, the years of the Indo-Soviet Treaty of Peace, Friendship and Cooperation and of the American tilt towards Pakistan during the Bangladesh war (Raghavan 2013). It returns close to zero in 1978, under the Janata government that had promised a "genuine" non-alignment, and drops back to between -0.5 and -0.7 in the 1980s, after the return of Indira Gandhi and India's refusal to condemn the Soviet intervention in Afghanistan. None of these phases can be read from the Alignment Index, which is close to zero and noisy until the late 1970s and then falls towards -1 in the 1980s, as it does, to varying degrees, for every other country in the sample.

[8] See Pant and Super (2015).

4.1.2 USA–China Rivalry

The rise of China has increasingly challenged the USA's hegemonic position, marking a shift in the international order following the end of the Cold War.[9] The emergence of two competing centres of geopolitical influence has led countries to reconsider their international positioning, in some respects resembling the dynamics of the Cold War, albeit in a substantially different economic and institutional context.[10]

Examining countries' relative alignment with the USA and China therefore provides a useful opportunity to assess the informational content of the proposed HI relative to the conventional Alignment Index. The comparison is of particular interest because the two poles occupy very different places in the distribution of ideal points: the USA sits at one extreme, China close to the large group of developing countries. Figures 6–9 report the results.

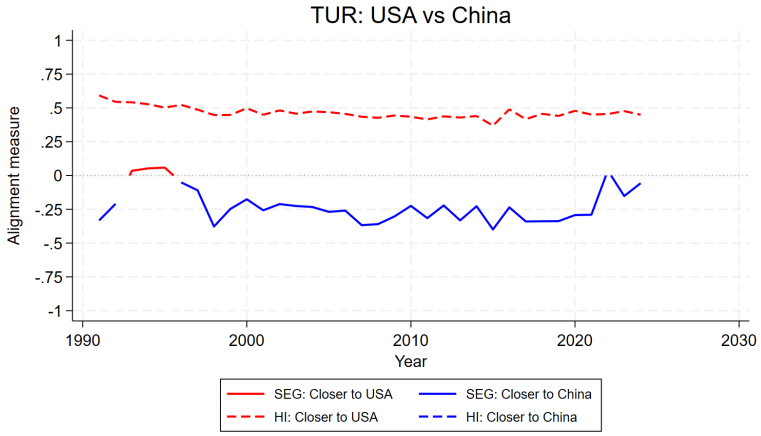


Figure 6: Türkiye: HI vs SEG during the USA–China rivalry.

The Alignment Index in Figure 6 might suggest that Türkiye has aligned itself with China: it is negative in every year from 1996 to 2021, mostly between -0.2 and -0.4. The HI, however, reveals a different picture: Türkiye's voting patterns have remained consistently closer to those of the USA, with values between 0.4 and 0.6 and no trend. Two remarks are in order. First, the level of the HI is well below that of France and the United Kingdom, which is what one expects of a NATO member that pursues an increasingly autonomous foreign policy (Kutlay and Öniş 2021). Second, the HI does not react to the bilateral crises of the period, such as the refusal to open a northern front against Iraq in 2003 or the purchase of the Russian S-400 system and the ensuing American sanctions: these disputes were never put to a vote in the Assembly, and UNGA

[9] See Broner, Martin, Meyer, Trebesch, and Zhou Wu (2025) and Layne (2018).
[10] See Brands and Gaddis (2021).

data, whatever the index built on them, cannot capture them. The Alignment Index returns to zero only in 2022, when Türkiye voted with the West on the resolutions condemning the invasion of Ukraine.

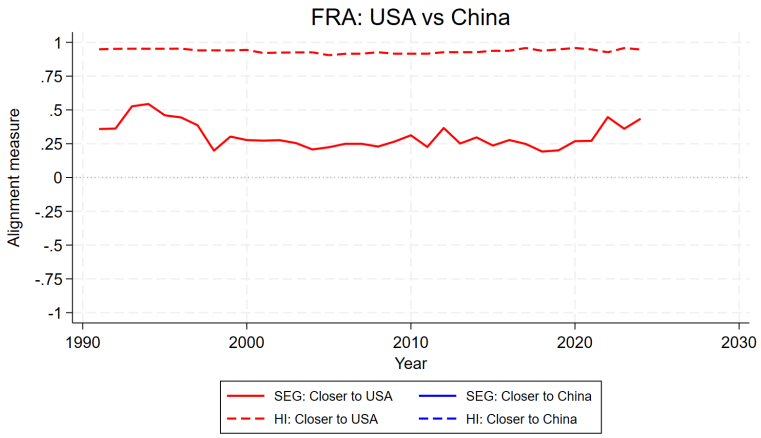


Figure 7: France: HI vs SEG during the USA–China rivalry.

In the post-Cold War period, France has remained a close ally of the USA. The HI in Figure 7 captures this pattern clearly, showing France's relative proximity to the American position throughout the period, at values between 0.9 and 0.96. The Alignment Index agrees on the sign but not on the intensity: it lies between 0.2 and 0.3 for most of the period, a level that would describe a lukewarm partner and not one of Washington's closest allies. The rift over the invasion of Iraq in 2003, deep enough to affect bilateral trade (Michaels and Zhi 2010), leaves only a faint trace in the HI, which touches its minimum of 0.9 in 2005.

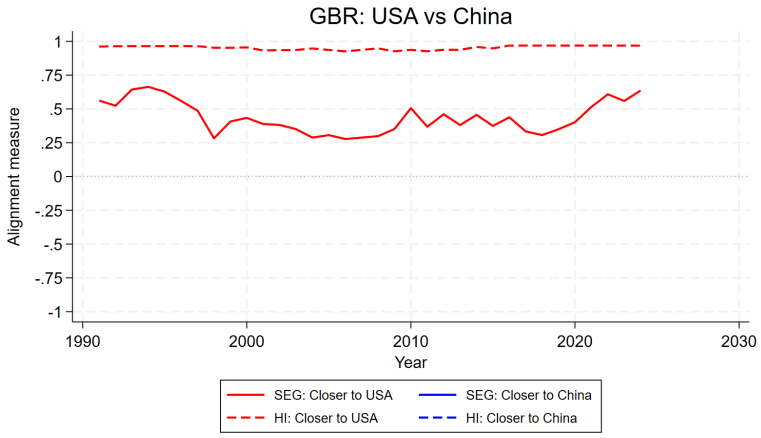


Figure 8: United Kingdom: HI vs SEG during the USA–China rivalry.

Similarly to France, the United Kingdom has remained closely aligned with the USA in its UNGA voting behaviour. The HI in Figure 8 makes this relative positioning particularly clear: it never falls below 0.92. The Alignment Index moves between 0.3 and 0.65 and, as for France, increases after 2020. In both countries the amplitude of the Alignment Index is several times that of the HI, although nothing in the diplomatic record of the period suggests that their position relative to Washington and Beijing changed to that extent.

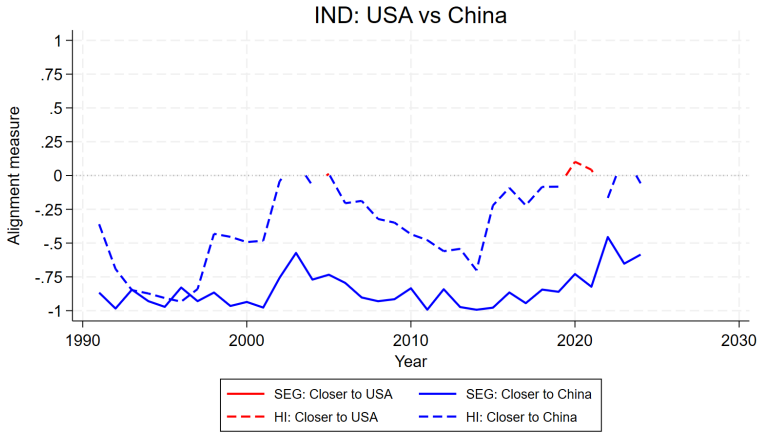


Figure 9: India: HI vs SEG during the USA–China rivalry.

India's non-alignment continued well beyond the end of the Cold War. Nevertheless, its voting behaviour reveals a closer relative alignment with China than with the USA for much of the period, although this pattern has changed

in more recent years. The complex dynamics of India's diplomatic positioning are not adequately captured by the conventional Alignment Index, which in Figure 9 lies between -0.55 and -1 throughout and below -0.7 in most years. A value of -1 denotes perfect alignment with China, which is hard to reconcile with a relationship marked by a border war, an unresolved territorial dispute and, in 2020, the deadliest clash between the two armies in decades.

The HI provides a clearer picture of India's relative position within the international distribution of UNGA voting patterns, and one that matches the evolution of its foreign policy described by Pant and Super (2015). The index is strongly negative in the mid-1990s; it rises after 1998 and reaches zero between 2002 and 2005, the years of the strategic partnership with Washington that led to the civil nuclear agreement of 2005. It declines again between 2006 and 2014, the period in which India coordinated with China in the BRICS and in the climate negotiations, and it jumps back towards zero from 2015. The only clearly positive values of the whole series are those of 2020 and 2021, immediately after the clash in the Galwan Valley and the revival of the Quad. In 2022–24 the index returns to slightly negative values, consistent with India's abstentions on Ukraine. India is, on this evidence, the one country in the sample that has genuinely moved between the two poles, and it is today approximately equidistant from them.

4.1.3 Summary of the Case Studies

Table 1 collects the evidence of Figures 1–9. Three regularities emerge. First, wherever the two indices disagree, the diplomatic record sides with the HI. Second, the errors of the Alignment Index are not idiosyncratic: it falls for all five countries in the 1980s, including the two closest allies of the USA, and it places the two countries of the sample outside the Western core closer to China than to the USA after 1991. Both patterns are what the second and third shortcomings listed above predict. Third, the HI separates countries that are stable (France and the United Kingdom after 1970, Türkiye after 1991) from countries that move (Türkiye in 1964–78, India throughout), a distinction that matters for the identification of non-aligned and connector countries in the Geoeconomics literature.

Table 1: Alignment Index (SEG) and Hegemon Index (HI) against the historical record

Country	SEG	HI	Historical record
Panel A: USA vs USSR, 1946–1990			
Türkiye	Near 1 until 1963; near 0 afterwards; closer to USSR after 1985	0.4–0.65 until 1963; trough in 1965–76; back to 0.7 in the 1980s	Johnson letter (1964), opening to Moscow, Cyprus and embargo (1974–78), realignment after 1980
France	Troughs in 1956 and 1967; falls to 0.45 by 1990	0.65–0.8 in 1961–68; above 0.9 from 1971	Suez, Gaullist decade, rapprochement in the 1970s and 1980s
United Kingdom	0.55–0.85 in the 1950s; falls to 0.55 by 1990	0.8–0.97 throughout; maximum in the late 1980s	Colonial disputes and Suez; closest partnership in the 1980s
Yugoslavia	Closer to USA in 1951–53; mildly closer to USSR afterwards	Closer to USSR throughout; Western tilt in 1952–53	Split with Stalin, Western aid, reconciliation of 1955, non-alignment
India	Equidistant until mid-1970s; at times closer to USA	Closer to USSR throughout; near 0 in 1965 and 1978; −0.7 in 1971–72	Pakistan pact (1954), war with China (1962), Indo-Soviet treaty (1971), Janata government (1977–79)
Panel B: USA vs China, 1991–2024			
Türkiye	Closer to China in 1996–2021	0.4–0.6, closer to USA, no trend	NATO member with growing strategic autonomy
France	0.2–0.55	0.9–0.96	Close ally; Iraq rift in 2003
United Kingdom	0.3–0.65	0.92–0.97	Close ally
India	−0.55 to −1	Near 0 in 2002–05; −0.7 in 2014; positive in 2020–21	Nuclear agreement (2005), BRICS, Galwan clash and Quad (2020)

Values are approximate readings of Figures 1–9. Positive values denote a position closer to the USA.

4.2 Geoeconomic Vulnerability: A New Perspective

4.2.1 Recomputing GeoV

One of the most relevant relationships under scrutiny in Geoeconomics is that between trade flows and geopolitical distance. Aiyar and Ohnsorge (2024), for example, follow a straightforward intuition: the more ideologically distant a country is from its trading partners, the more vulnerable it will be to geopolitical tensions. They therefore propose to measure Geoeconomic Vulnerability as a weighted average of a country i's Ideal Point Distances, where weights are determined by i's exports or imports.

Employing IPD directly is, as discussed in Section 4.1, subject to several drawbacks: being a dimensionless measure, it is hard to compare across time and countries. AP can thus be used to rewrite[11]

$$AP\text{-}GeoV_{it} = \sum_{j \in J} \frac{y_{ijt}}{\sum_j y_{ijt}} (1 - AP_{ijt}), \tag{3}$$

where y_{ijt} is a trade flow between country i and partner j and AP_{ijt} stands for either of the two percentiles that can be attached to the pair, as discussed below. Since percentiles lie between zero and one, $AP\text{-}GeoV_{it}$ has a natural benchmark: a country whose trade were unrelated to geopolitical rankings would score about 0.5, values below 0.5 indicate trade that is tilted towards closely ranked partners, and values above 0.5 indicate trade tilted towards distant ones. No such benchmark exists for a weighted average of distances.

An interesting feature of AP is that it is nonsymmetric, unlike IPD, so that (3) can be computed from two different perspectives. With the convention of Section 4.1, AP_{ijt} is the position of partner j in the ordering of country i: it measures how i ranks the countries it trades with, and I call this the *outward ranking* of i. Conversely, AP_{jit} is the position of i in the ordering of partner j: it measures how i is ranked by the countries it trades with, and I call the resulting measure the *inward ranking* of i.

Given this distinction, and the choice between export and import weights, $AP\text{-}GeoV_{it}$ can be calculated in four ways. In the acronyms that follow, the first letter refers to the weights (X for exports, M for imports) and the second to the ranking perspective: O for outward, that is, how country i ranks its trading partners, and I for inward, that is, how its trading partners rank country i.

- With export weights and the outward ranking:

$$XOAP\text{-}GeoV_{it} = \sum_{j \in J} \frac{x_{ijt}}{\sum_j x_{ijt}} (1 - AP_{ijt})$$

- With export weights and the inward ranking:

$$XIAP\text{-}GeoV_{it} = \sum_{j \in J} \frac{x_{ijt}}{\sum_j x_{ijt}} (1 - AP_{jit})$$

- With import weights and the outward ranking:

$$MOAP\text{-}GeoV_{it} = \sum_{j \in J} \frac{m_{ijt}}{\sum_j m_{ijt}} (1 - AP_{ijt})$$

- With import weights and the inward ranking:

$$MIAP\text{-}GeoV_{it} = \sum_{j \in J} \frac{m_{ijt}}{\sum_j m_{ijt}} (1 - AP_{jit})$$

[11] I take the complement of the Alignment Percentile, $1 - AP$, to preserve the analogy with IPD: a higher value denotes a more distant partner.

Here x_{ijt} denotes exports of i to j and m_{ijt} imports of i from j. Table 2 summarises the four measures. $XOAP$ and $MOAP$ use country i's own ordering, respectively of its customers and of its suppliers, and are therefore own-ranking exposures; $XIAP$ and $MIAP$ use the ordering of the partners and are partner-ranking exposures.

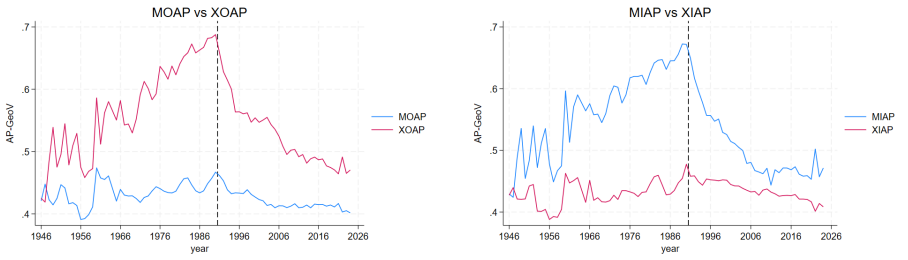
Table 2: The four AP-based measures of geoeconomic vulnerability

Measure	Trade weights	Percentile	Interpretation for country i
$XOAP$	Exports of i	AP_{ijt}	How far i ranks its destinations (outward ranking)
$MOAP$	Imports of i	AP_{ijt}	How far i ranks its suppliers (outward ranking)
$XIAP$	Exports of i	AP_{jit}	How far its customers rank i (inward ranking)
$MIAP$	Imports of i	AP_{jit}	How far its suppliers rank i (inward ranking)

AP_{ijt} is the percentile of j in the ordering of i (outward ranking); AP_{jit} is the percentile of i in the ordering of j (inward ranking). All measures are trade-weighted averages of $1 - AP$ and lie between 0 and 1.

We can therefore exploit the nonsymmetric nature of ranking percentiles in order to understand two different dimensions of geopolitical risk: one related to a country's ideological preferences regarding its trading partners and the other one concerning the trading partners' ideological preferences regarding the country. The distinction has a direct counterpart in the theory of economic coercion. Own-ranking exposure measures how much of its trade a country carries out with partners it would itself regard as adversaries, which is the exposure relevant for a country that contemplates sanctions or de-risking. Partner-ranking exposure measures how much of its trade depends on partners that regard it as an adversary, which is the exposure relevant for a country that fears being the target of coercion (Hirschman 1945; Clayton, Maggiori, and Schreger 2024).

4.2.2 Aggregate Analysis



(a) $MOAP$ and $XOAP$ (b) $MIAP$ and $XIAP$

Figure 10: AP-based geoeconomic vulnerability, world average, 1946–2024. The vertical line marks the end of the Cold War.

Figure 10 reports the world average of the four measures. An interesting feature of the results is that the four series form two pairs that cut across the distinction between outward and inward rankings. $XOAP$ and $MIAP$ display remarkably similar trajectories, as do $XIAP$ and $MOAP$. This is not the grouping one would expect from the definitions alone: the first pair combines an outward measure based on exports with an inward measure based on imports, while the second combines an inward measure based on exports with an outward measure based on imports.

The distinction is important for interpretation. Outward measures capture the geopolitical distance of a country's trading partners as assessed by the country itself, whereas inward measures capture the distance assigned to the country by its trading partners. The fact that measures based on opposite ranking perspectives nevertheless move together suggests that the geopolitical structure of trade cannot be understood solely by looking at how countries rank their partners or how partners rank them. The direction of trade also matters, and the relationship between the two perspectives appears to be systematically associated with it.

The first pair, $XOAP$ and $MIAP$, starts close to 0.42 in 1946, rises almost without interruption to a peak of 0.67–0.69 in 1989–90, and then declines steeply after the end of the Cold War, reaching about 0.47 in recent years. The second pair, $XIAP$ and $MOAP$, presents a much more stable trajectory: the two series fluctuate between 0.39 and 0.48 over the period, with a slight decline beginning around the end of the Cold War, from about 0.47 in 1990 to about 0.40 in 2024.

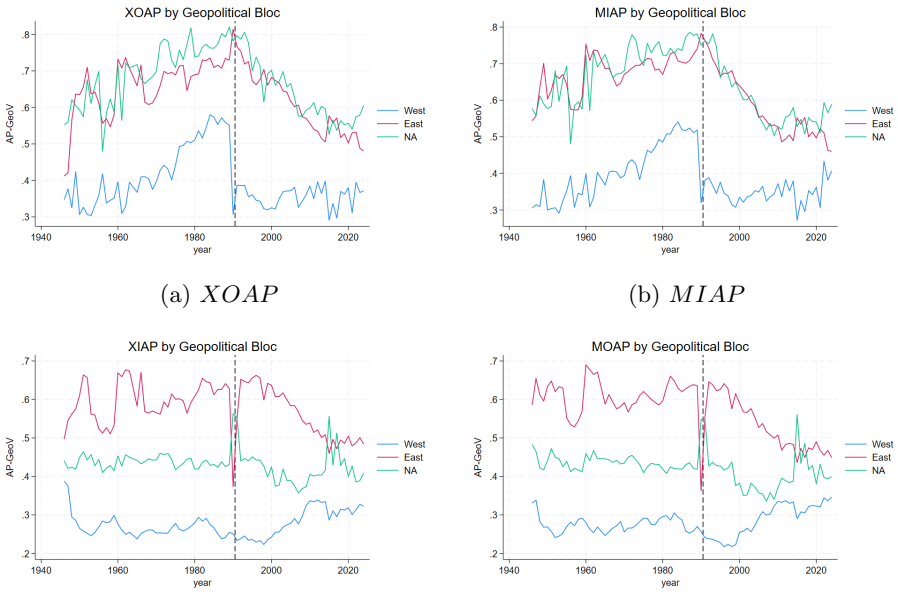
The contrast between these two empirical patterns is particularly interesting because it does not correspond to a simple distinction between own-ranking and partner-ranking exposure. Rather, it indicates that the relationship between the two ranking perspectives changes depending on whether trade is measured through exports or imports. In particular, the strong common movement of $XOAP$ and $MIAP$ suggests that a country's assessment of its export destinations and the assessment of that country by its import suppliers are closely associated at the aggregate level. The corresponding relationship between $XIAP$ and $MOAP$ is also visible, but follows a substantially more stable trajectory.

The historical evolution of the series suggests an explanation for these patterns. Membership of the UN rose from 51 states in 1945 to 99 in 1960 and 159 in 1990, almost entirely through decolonisation; the jump visible in all four series in 1960 coincides with the admission of seventeen new members in that single year. The new majority organised itself in the Non-Aligned Movement and in the Group of 77 and voted increasingly far from the Western countries, which nevertheless remained the main markets and the main suppliers of the developing world (Westad 2005; Dinkel 2019). The growing ideological distance between the new majority and the Western countries therefore affected both sides of trade relationships, although not necessarily symmetrically. The divergence between the two empirical pairs suggests that the consequences of this transformation depended on the direction in which trade relationships and geopolitical rankings were combined.

The fall after 1990 is consistent with two developments. The first is the

ideological rapprochement to the West by countries that were formerly part of the Soviet bloc, together with the reorientation of their trade towards Western Europe:[12] partners that had been at the far end of their ordering moved to the near end. The second is the growth of trade among developing countries, and with China in particular, which altered the composition of trading relationships and the geopolitical positions of the countries involved. These developments affected both outward and inward rankings, but the fact that $XOAP$ and $MIAP$ declined together while $XIAP$ and $MOAP$ remained comparatively stable indicates that the adjustment was not uniform across the four measures.

In recent years the two series in the first pair have stabilised, with an isolated spike in 2022. The second pair has remained comparatively stable throughout the post-Cold War period. These results provide a relevant qualification to the current debate on fragmentation: trading with geopolitically distant partners is not a novelty of the present decade, and the world economy sustained a much larger degree of geopolitical exposure throughout the Cold War. At the same time, the crossed correlation structure cautions against treating geoeconomic vulnerability as a single dimension. Whether geopolitical distance is assessed by the trading country or by its partners, and whether exposure is measured through exports or imports, makes a substantive difference.



(a) $XOAP$ (b) $MIAP$

(c) $XIAP$ (d) $MOAP$

Figure 11: AP-based geoeconomic vulnerability by geopolitical bloc, 1946–2024. The panels show the four measures separately. NA denotes non-aligned countries.

[12] See, for example, Aslund (2012).

Figure 11 decomposes the four measures by geopolitical bloc.

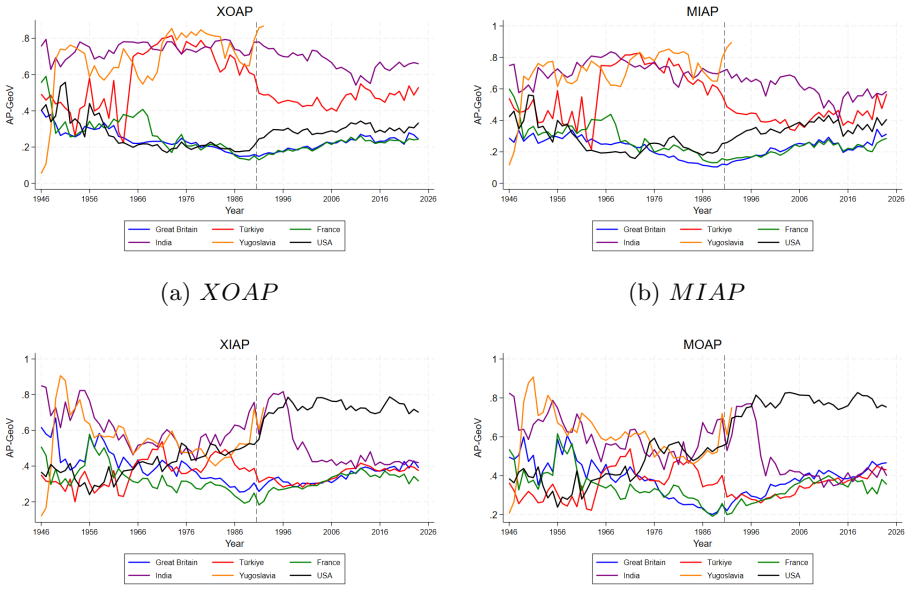
The decomposition reveals substantial differences across geopolitical blocs. The Eastern and non-aligned countries display the highest levels of exposure on several measures, with some series reaching about 0.8 around 1990 and subsequently declining to approximately 0.5–0.6. The Western bloc is generally less exposed, although the magnitude of this advantage varies across the four measures. These differences are consistent with the historical structure of international trade: Western countries traded predominantly with one another, while Eastern and non-aligned countries maintained substantial commercial relationships with countries located at different positions in their respective geopolitical rankings.

The bloc-level results also help clarify the crossed correlation observed in the world averages. The two series that move together globally, $XOAP$ and $MIAP$, do not measure the same ranking perspective, but their similar trajectories suggest that the historical transformations of trade and geopolitical alignment affected both outward assessments of export destinations and inward assessments of import suppliers. Similarly, the comparatively stable behaviour of $XIAP$ and $MOAP$ indicates that the relationship between these two dimensions evolved differently. The bloc decomposition makes it possible to examine whether these aggregate patterns reflect common movements across geopolitical groups or the offsetting behaviour of countries belonging to different blocs.

The West is generally less vulnerable geoeconomically than the other blocs, although this conclusion must be qualified by the distinction between outward and inward exposure. Western countries' own assessments of their trading partners are generally consistent with their dense commercial relationships within the Western bloc. Their partners' assessments, however, need not coincide with this picture. The distinction is particularly relevant for understanding the position of the United States, whose international political standing cannot necessarily be inferred from the relatively close ideological relationships among Western European countries.

Two qualifications are necessary. First, the Western advantage has been shrinking on several measures. The partner-ranking exposures of the West reach their minimum in the second half of the 1990s and have increased since, from about 0.22 to about 0.35, over the period in which China joined the WTO and became a major supplier and customer of the advanced economies. Over the same years, the corresponding measures for the Eastern bloc decline, so that the three groups are closer to each other today than at any time since 1946. Second, the relative stability of some world averages does not imply an absence of change at the bloc level: different trajectories across geopolitical groups can offset one another in the aggregate.

4.2.3 Country Case Studies



(a) $XOAP$ (b) $MIAP$

(c) $XIAP$ (d) $MOAP$

Figure 12: AP-based geoeconomic vulnerability by country, 1946–2024. The panels show the four measures separately.

Figure 12 reports the four measures for the five countries of Section 4.1 and for the USA. The country series connect the two parts of the paper, since they show how the diplomatic turning points identified by the HI translate into economic exposure.

The country-level evidence illustrates why the distinction between outward and inward rankings matters. The two perspectives need not yield the same assessment of a country's geopolitical exposure: a country may rank its trading partners relatively close while those same partners rank it as relatively distant. Moreover, the crossed correlation observed in the world averages suggests that these differences are systematically related to the direction of trade.

Consider first the outward measures, $XOAP$ and $MOAP$. Yugoslavia moves from the lowest values of the sample in 1946 to about 0.7 by 1950: after the split of 1948 and the economic blockade imposed by the Soviet bloc, its trade was redirected towards the West (Lees 1997), while its votes, as shown in Figure 4, remained far from the Western ones. Türkiye displays a jump of similar size in the mid-1960s, from about 0.4 to between 0.7 and 0.8, exactly when the HI in Figure 1 collapses: its votes moved away from those of its Western partners, its trade did not. The exposure returns to about 0.5 around 1990. India, whose commerce was oriented towards countries it voted against, stays between 0.7 and

0.8 until the end of the Cold War and declines gradually afterwards. France, the United Kingdom and the USA remain below 0.4 from the 1970s onwards.

The inward measures, $XIAP$ and $MIAP$, provide a complementary perspective. They capture the extent to which a country's trading partners regard it as geopolitically distant. The USA displays the most striking result: while its inward exposure was at or below the level of the European countries until the 1960s, it rises to between 0.7 and 0.8 in the first half of the 1990s and has remained there since, far above every other country in the figure. The USA ranks its trading partners as relatively close, but its trading partners, including many allies, rank the USA among the most distant members of the Assembly, where Washington has often voted in a very small minority since the end of the Cold War (Voeten 2000). The statement that the West is less vulnerable than commonly perceived therefore applies to Western Europe and does not extend to the hegemon when vulnerability is measured from the partners' side.

India moves in the opposite direction: its inward exposure falls from about 0.8 in the mid-1990s to about 0.4 at the turn of the century, the years in which the HI of Figure 9 records its approach to the USA, and has been stable since.

Taken together, the country cases reinforce the central point of the section. Outward and inward exposure provide distinct information about geopolitical vulnerability, and their relationship depends on the direction in which trade is measured. The fact that $XOAP$ and $MIAP$ tend to move together, as do $XIAP$ and $MOAP$, is therefore not a reason to collapse the four measures into two equivalent concepts. Rather, it highlights an empirical regularity that deserves further investigation and shows why a multidimensional approach to geoeconomic vulnerability is useful.

4.3 The Gravity of Reciprocity

4.3.1 Specification

An important topic in Geoeconomics is understanding whether geopolitical alignment influences trade. However, there has been no focus on understanding the impact of geopolitical reciprocity on international trade, although reciprocity is regarded in International Relations as a basic condition for cooperation (Keohane 1986) and asymmetry as a source of power (Hirschman 1945). I propose to apply the gravity framework to exploit the nonsymmetric nature of AP. I therefore construct the following measure:

$$R_{ijt} = \frac{AP_{ijt} - AP_{jit}}{AP_{ijt} + AP_{jit}} \tag{4}$$

R_{ijt} is bounded in $[-1; 1]$ and provides information on whether the country pair ij entertains a geopolitically balanced relationship. If $R_{ijt} = 0$ the two countries place each other at the same percentile of their respective orderings. If $R_{ijt} > 0$ (resp. < 0), then j is relatively closer to (resp. further from) i in the ordering of i than i is in the ordering of j: the affinity of i for j exceeds (resp. falls short of) the affinity of j for i. Note that $R_{ijt} = -R_{jit}$, so that R^2_{ijt}

is common to the two directions of trade within a pair and measures the degree of imbalance irrespective of which country is the more favourably disposed. I run the following Poisson Pseudo Maximum Likelihood (henceforth PPML) regression (Santos Silva and Tenreyro 2006):

$$y_{ijt} = \exp\Big(\sum_{p} \mathbf{1}[t \in p]\left(\beta_{1p} R_{ijt} + \beta_{2p} R_{ijt}^2\right) + \gamma_{it} + \delta_{jt} + \eta_{ij}\Big)\,\varepsilon_{ijt}, \qquad (5)$$

where y_{ijt} is a trade flow from i to j at time t, p indexes sub-periods, and γ_{it}, δ_{jt} and η_{ij} are exporter-year, importer-year and pair fixed effects. Standard errors are clustered by ij pairs. The exporter-year and importer-year effects absorb the multilateral resistance terms of structural gravity (Anderson and Wincoop 2003) together with any country-specific shock, including the overall geopolitical position of each country in a given year. The pair effects absorb all time-invariant determinants of bilateral trade, such as distance, common language and colonial ties. The coefficients are therefore identified by changes over time in the balance of the relationship within a pair, relative to the average level of the pair. The choice of a quadratic form for R_{ijt} fits the way this reciprocity measure is constructed. That is, I expect that too extreme values of R_{ijt} have symmetric effects on trade: β_{2p} captures the effect of imbalance as such, and β_{1p} any difference between exporting to a partner that is less, or more, favourably disposed than oneself.

4.3.2 Results

Table 3 reports the estimates for three sub-periods: the Cold War, the decade that followed it, and the years since 2002. Table 4 splits the last period into the years before the Global Financial Crisis, those between the crisis and the invasion of Ukraine, and 2022–24.

Table 3: PPML estimates of the effect of reciprocity R_{ij} on trade flows: three sub-periods

	1948-1991	1992-2001	2002-2024
R_{ij}	0.038	-0.056**	0.019
	(0.035)	(0.023)	(0.014)
R_{ij}^2	-0.061	-0.261***	0.063***
	(0.048)	(0.046)	(0.018)
Observations		1,181,181	
Origin × Year FE		Yes	
Destination × Year FE		Yes	
Origin–Destination Pair FE		Yes	
Clusters		32,617	
Pseudo R^2		0.990	

Standard errors clustered by origin–destination pair in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: PPML estimates of the effect of reciprocity R_{ij} on trade flows: five sub-periods

	1948-1991	1992-2001	2002-2008	2009-2021	2022-2024
R_{ij}	0.037	-0.057**	-0.042**	0.033**	0.060
	(0.035)	(0.023)	(0.020)	(0.017)	(0.045)
R_{ij}^2	-0.061	-0.261***	-0.146***	0.048***	0.429***
	(0.048)	(0.047)	(0.031)	(0.018)	(0.084)
Observations			1,181,181		
Origin × Year FE			Yes		
Destination × Year FE			Yes		
Origin–Destination Pair FE			Yes		
Clusters			32,617		
Pseudo R^2			0.990		

Standard errors clustered by origin–destination pair in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Three results stand out.

The Cold War. Neither coefficient is statistically different from zero between 1948 and 1991 in either table. During the bipolar period, the balance of a political relationship had no detectable association with trade once pair and country-year effects are controlled for. This is in line with the evidence that trade was then governed by bloc membership (Campos, Heid, and Timini 2024; Gokmen 2017): between blocs it was restricted by export controls and state trading, within blocs it was organised by institutions such as the CMEA, and in both cases marginal changes in voting affinity had little room to matter.

The unipolar years. Between 1992 and 2001 the coefficient on R_{ij}^2 is -0.261 and highly significant in both tables: the more unbalanced a relationship became, the less the pair traded. The implied effect is not negligible. Relative to a perfectly balanced relationship, an imbalance of $|R_{ij}| = 0.5$ is associated with trade that is about 6% lower through the quadratic term alone, and the extreme case $|R_{ij}| = 1$ with trade about 23% lower. The linear coefficient is negative as well (-0.056), which makes the penalty larger for the exporter whose affinity is not returned: at $R_{ij} = 0.5$ exports are about 9% lower, at $R_{ij} = -0.5$ about 4% lower. The estimated relationship peaks at $R_{ij} \approx -0.11$, close to zero, as the symmetric specification presumes. The penalty for imbalance is still present, at roughly half the size, in 2002–08 (-0.146 in Table 4). In the decade and a half in which a single hegemon promoted common rules and the multilateral trading system expanded, with the creation of the WTO and the enlargement of the European Union, trade grew among countries whose political affinity was mutual. This is what a theory of hegemonic globalisation driven by policy alignment predicts (Broner, Martin, Meyer, and Trebesch 2025).

The reversal. After 2008 the sign changes. The coefficient on R_{ij}^2 is small and positive in 2009–21 (0.048) and large and positive in 2022–24 (0.429): in the most recent years an imbalance of $|R_{ij}| = 0.5$ is associated with trade about 11% higher than in a balanced relationship. Table 3 shows why the finer partition

matters: the pooled coefficient for 2002–24 (0.063) averages a negative, a mildly positive and a strongly positive sub-period, and would by itself suggest a modest and stable premium.

This reversal is the empirical puzzle that the paper puts forward. Standard reasoning, and the evidence of the 1990s, suggest that one-sided relationships should be less conducive to trade than reciprocated ones. I see three candidate explanations, which are not mutually exclusive. The first is the emergence of connector countries. Imbalance is largest in pairs formed by a country at one end of the ideal-point line and a country in its crowded centre, which is where the connectors identified by Aiyar and Ohnsorge (2024) and Gopinath et al. (2025) are located; if trade between the blocs is being rerouted through them, it grows precisely in unbalanced pairs. The second is the rise of China as the main trading partner of a large number of countries on both sides of the political spectrum, a process that accelerated after 2008 and that links trade growth to pairs in which the political relationship is not reciprocal. The third is friend-shoring by the hegemons themselves: as rivalry intensifies, each pole deepens its trade with countries it is courting, that is, with partners it ranks more favourably than they rank it.

The estimates for 2022–24 call for caution. They rest on three years of data, in which the votes of the emergency special session on Ukraine reshuffled the orderings and commodity prices moved sharply; the country-year effects absorb the latter but not changes in the composition of bilateral trade. The estimates are, moreover, conditional associations and not causal effects, since the balance of a political relationship may itself respond to trade. Even with these caveats, the pattern across periods is systematic: the coefficient on R^2_{ij} increases monotonically from the 1990s to the present, and it does so in a specification that absorbs every country-level and every time-invariant bilateral determinant of trade. A symmetric measure of geopolitical distance cannot detect this pattern, because the regressor that reveals it does not exist when $AP_{ijt} = AP_{jit}$.

5 Conclusion

This paper proposes a simple measure of geopolitical alignment, the Alignment Percentile, that overcomes the flaws of existing measurements based on ideal point distances. Replacing distances with ranks makes alignment comparable over time, anchors it to the distribution of all countries in a given year and, above all, makes it directional: the way a country regards a partner and the way that partner regards the country become two distinct objects.

The findings are relevant for three audiences. For researchers who classify countries into blocs, non-aligned and connectors, the two case studies show that the index currently in use can mistake the isolation of a hegemon for the drift of its allies: it describes the United Kingdom and France as moving away from the United States in the 1980s and Türkiye and India as aligned with China, while the percentile-based index recovers the turning points documented by diplomatic historians. Classifications that feed into quantitative fragmentation

scenarios should be re-examined in this light. For the measurement of geoeconomic vulnerability, the distinction between own-ranking and partner-ranking exposure shows that the geopolitical exposure of world trade peaked in 1990 and is now near its post-war minimum, that Western Europe has always been the least exposed region, and that the United States is the country whose trading partners regard it as most distant. For the study of trade, the gravity estimates show that the reciprocity of a political relationship mattered in the 1990s in the way theory would suggest, and that the relationship has reversed in the years of renewed great-power rivalry.

Several extensions are left for future work. The first is to assess whether Alignment Percentiles vary significantly when computed on subsets of UNGA votes, such as economic resolutions only, and when they are derived from sources other than the Assembly, such as treaties. The second is to discriminate between the explanations of the reciprocity puzzle, using sectoral trade data and an explicit identification of connector countries. The third is to extend the analysis from trade to investment and financial flows, where the asymmetry between the views of the source and of the destination country is likely to be at least as relevant.

References

Airaudo, Florencia S. et al. (Mar. 2025). "Fragmentation? Revisiting the Ideal Point Distance Measure of Geopolitical Distance". In: *FEDS Notes*. DOI: 10.17016/2380-7172.3749. URL: https://doi.org/10.17016/2380-7172.3749.

Aiyar, Shekhar and Franziska Ohnsorge (Aug. 2024). *Geoeconomic Fragmentation and "Connector" Countries*. MPRA Paper 121726. University Library of Munich, Germany. URL: https://ideas.repec.org/p/pra/mprapa/121726.html.

Anderson, James E. and Eric van Wincoop (2003). "Gravity with Gravitas: A Solution to the Border Puzzle". In: *American Economic Review* 93.1, pp. 170–192.

Aslund, Anders (2012). *How Capitalism Was Built: The Transformation of Central and Eastern Europe, Russia, the Caucasus, and Central Asia*. 2nd ed. Cambridge University Press.

Bailey, Michael A., Anton Strezhnev, and Erik Voeten (2017). "Estimating Dynamic State Preferences from United Nations Voting Data". In: *The Journal of Conflict Resolution* 61.2, pp. 430–456. URL: http://www.jstor.org/stable/26363889.

Baur, Andreas et al. (2025). "Geoeconomic fragmentation and the role of non-aligned countries". In: *Economics Letters* 255, p. 112533. ISSN: 0165-1765. DOI: https://doi.org/10.1016/j.econlet.2025.112533. URL: https://www.sciencedirect.com/science/article/pii/S0165176525003702.

Bonadio, Barthélémy et al. (Oct. 2024). "Playing with Blocs: Quantifying Decoupling". Working Paper. URL: https://bbonadio.github.io/BHKLPTT_PlayingWithBlocs.pdf.

Bozo, Frédéric (2001). *Two Strategies for Europe: De Gaulle, the United States, and the Atlantic Alliance*. Lanham: Rowman & Littlefield.

Brands, Hal and John Lewis Gaddis (2021). "The New Cold War: America, China, and the Echoes of History". In: *Foreign Affairs* 100.6, pp. 10–21. ISSN: 00157120, 23277793. URL: https://www.jstor.org/stable/27121438 (visited on 02/08/2026).

Broner, Fernando, Alberto Martin, Josefin Meyer, and Christoph Trebesch (2025). *Hegemonic Globalization*. CESifo Working Paper Series 11951. CESifo. URL: https://ideas.repec.org/p/ces/ceswps/_11951.html.

Broner, Fernando, Alberto Martin, Josefin Meyer, Christoph Trebesch, and Jiaxian Zhou Wu (May 2025). "Hegemony and International Alignment". In: *AEA Papers and Proceedings* 115, pp. 593–98. DOI: 10.1257/pandp.20251041. URL: https://www.aeaweb.org/articles?id=10.1257/pandp.20251041.

Campos, Rodolfo G., Benedikt Heid, and Jacopo Timini (2024). *The economic consequences of geopolitical fragmentation: Evidence from the Cold War*. arXiv: 2404.03508 [econ.GN].

Clayton, Christopher, Matteo Maggiori, and Jesse Schreger (Dec. 2024). *A Theory of Economic Coercion and Fragmentation*. Working Paper 33309. National Bureau of Economic Research. DOI: 10.3386/w33309. URL: http://www.nber.org/papers/w33309.

Dinkel, J. (2019). *The Non-Aligned Movement: Genesis, Organization and Politics (1927-1992)*. Trans. by A. Skinner. Brill.

Dumbrell, John (2006). *A Special Relationship: Anglo-American Relations from the Cold War to Iraq*. 2nd ed. Basingstoke: Palgrave Macmillan.

Farrell, Henry and Abraham L. Newman (2019). "Weaponized Interdependence: How Global Economic Networks Shape State Coercion". In: *International Security* 44.1, pp. 42–79.

Fouquin, Michel and Jules Hugot (2016). *Two Centuries of Bilateral Trade and Gravity Data: 1827–2014*. CEPII Working Paper 2016-14. CEPII. URL: https://www.cepii.fr/PDF_PUB/wp/2016/wp2016-14.pdf.

Gaddis, John Lewis (2005). *The Cold War: A New History*. New York: Penguin Press.

Gaulier, Guillaume and Soledad Zignago (Oct. 2010). *BACI: International Trade Database at the Product-Level. The 1994-2007 Version*. Working Papers 2010-23. CEPII. URL: https://www.cepii.fr/CEPII/fr/publications/wp/abstract.asp?NoDoc=2726.

Gokmen, Gunes (2017). "Clash of civilizations and the impact of cultural differences on trade". In: *Journal of Development Economics* 127.C, pp. 449–458. DOI: 10.1016/j.jdeveco.2016.12.008. URL: https://ideas.repec.org/a/eee/deveco/v127y2017icp449-458.html.

Gopinath, Gita et al. (2025). "Changing Global Linkages: A New Cold War?" In: *Journal of International Economics* 153, p. 104042. DOI: 10.1016/j.

jinteco.2024.104042. URL: https://www.sciencedirect.com/science/article/pii/S0022199624001697.

Gowa, Joanne and Edward D. Mansfield (1993). "Power Politics and International Trade". In: *American Political Science Review* 87.2, pp. 408–420.

Gulmez, Didem Buhari (2020). "The Resilience of the US–Turkey Alliance: Divergent Threat Perceptions and Worldviews". In: *Contemporary Politics* 26.4, pp. 475–492. DOI: 10.1080/13569775.2020.1777038.

Hale, William (2012). *Turkish Foreign Policy since 1774*. 3rd ed. Routledge. DOI: 10.4324/9780203102022.

Hanhimäki, Jussi M (2015). *The United Nations: A very short introduction*. Oxford University Press.

Hirschman, Albert O. (1945). *National Power and the Structure of Foreign Trade*. Berkeley: University of California Press.

Keohane, Robert O. (1986). "Reciprocity in International Relations". In: *International Organization* 40.1, pp. 1–27.

Kim, Soo Yeon and Bruce Russett (1996). "The New Politics of Voting Alignments in the United Nations General Assembly". In: *International Organization* 50.4, pp. 629–652.

Kutlay, Mustafa and Ziya Öniş (2021). "Turkish Foreign Policy in a Post-Western Order: Strategic Autonomy or New Forms of Dependence?" In: *International Affairs* 97.4, pp. 1085–1104.

Layne, Christopher (2018). "The US–Chinese power shift and the end of the Pax Americana". In: *International affairs* 94.1, pp. 89–111.

Lees, Lorraine M. (1997). *Keeping Tito Afloat: The United States, Yugoslavia, and the Cold War*. University Park: Pennsylvania State University Press.

Louis, Wm. Roger and Ronald Robinson (1994). "The Imperialism of Decolonization". In: *The Journal of Imperial and Commonwealth History* 22.3, pp. 462–511.

Luard, Evan (1982). *A History of the United Nations: Volume 1: The Years of Western Domination, 1945-1955*. Springer.

Lupu, Yonatan (2013). "The informative power of treaty commitment: Using the spatial model to address selection effects". In: *American Journal of Political Science* 57.4, pp. 912–925.

McMahon, Robert J. (1994). *The Cold War on the Periphery: The United States, India, and Pakistan*. New York: Columbia University Press.

Michaels, Guy and Xiaojia Zhi (2010). "Freedom fries". In: *American Economic Journal: Applied Economics* 2.3, pp. 256–281.

Moynihan, Daniel Patrick and Suzanne Weaver (1978). *A Dangerous Place*. Boston: Little, Brown.

Oğuzlu, H. Tarık (2023). "Turkey as a restrained middle power". In: *Turkish Studies* 24.3-4, pp. 673–690. DOI: 10.1080/14683849.2023.2171870.

Pant, Harsh V. and Julie M. Super (July 2015). "India's 'non-alignment' conundrum: a twentieth-century policy in a changing world". In: *International Affairs* 91.4, pp. 747–764. ISSN: 0020-5850. DOI: 10.1111/1468-2346.12336. eprint: https://academic.oup.com/ia/article-pdf/91/4/

747/13656806/inta12336.pdf. URL: https://doi.org/10.1111/1468-2346.12336.

Pollins, Brian M. (1989). "Does Trade Still Follow the Flag?" In: *American Political Science Review* 83.2, pp. 465–480.

Raghavan, Srinath (2013). *1971: A Global History of the Creation of Bangladesh*. Cambridge, MA: Harvard University Press.

Santos Silva, J. M. C. and Silvana Tenreyro (2006). "The Log of Gravity". In: *The Review of Economics and Statistics* 88.4, pp. 641–658.

Simmons, Beth A and Daniel J Hopkins (2005). "The constraining power of international treaties: Theory and methods". In: *American Political Science Review* 99.4, pp. 623–631.

Statler, Kathryn C. (Sept. 2016). *France-US Relations*. DOI: 10.1093/acrefore/9780199329175.013.372. URL: https://oxfordre.com/americanhistory/view/10.1093/acrefore/9780199329175.001.0001/acrefore-9780199329175-e-372.

Voeten, Erik (2000). "Clashes in the Assembly". In: *International Organization* 54.2, pp. 185–215.

Westad, Odd Arne (2005). *The Global Cold War: Third World Interventions and the Making of Our Times*. Cambridge: Cambridge University Press.